

① Find the z-scores for your marks on MCQ1, MCQ2 & MCQ3, treating the corresponding results as raw data.

$$MCQ1 = [8, 11, 8, 12, 9, 11, 15, 9, 9, 9]$$

$$2 = [15.5, 13.5, 14.5, 8, 15, 9.5, 16.5, 14.5, 14]$$

$$3 = [9, 8, 7, 11, 9, 9, 11, 10.5, 10, 7, 5]$$

$$Z = \frac{X - \mu}{\sigma}$$

for MCQ1

$$\mu = \frac{8 + 11 + 8 + 12 + 9 + 11 + 15 + 9 \times 3}{10} = 10.1$$

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x - \bar{x})^2}$$

$$= \sqrt{\frac{1}{10} [(8 - 10.1)^2 + (11 - 10.1)^2 + (8 - 10.1)^2 + (12 - 10.1)^2 + (9 - 10.1)^2 + (11 - 10.1)^2 + (15 - 10.1)^2 + (9 - 10.1)^2 \times 3]}$$

$$= 2.701$$

$$Z = \frac{X - \mu}{\sigma}$$

For, $X = 8$

$$Z = \frac{8 - 10.1}{2.701} = -1.04$$

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MCQ 2.

$$\bar{X} = \frac{16.5 + 18.5 + 14.5 + 8 + 15 + 9.5 + 16.5 + 14.5 + 6 + 14}{10}$$

$$= 12.8$$

$$\sigma = \sqrt{\frac{1}{10} \sum_{i=1}^{10} (x_i - \bar{x})^2}$$

$$= 3.466$$

For $x = 16.5$

$$Z = \frac{x - \mu}{\sigma} = \frac{16.5 - 12.8}{3.466}$$

$$= 1.068$$

MCQ 3

$$\mu = \frac{9 + 8 + 7 + 11 + 9 + 9 + 11 + 10.5 + 10 + 7 + 5}{11}$$

$$= 8.773$$

$$\sigma = \sqrt{\frac{1}{11} \sum_{i=1}^{11} (x_i - \bar{x})^2}$$

$$= 1.801$$

For, $x = 9$

$$Z = \frac{x - \mu}{\sigma} = \frac{9 - 8.773}{1.801} = 0.126$$

② Find the 40%, 60%, 80% percentile from MCQ 1, MCQ 2, MCQ 3 result

→ For 40% percentile. For MCQ 1 $n=10$.

$$i = \frac{P \cdot N}{100} = \frac{40 \times 10}{100} = 4$$

⑦ $M @ 1 = [8, 11, 8, 12, 9, 11, 15, 9, 9, 9]$

~~MCQ~~ For 60%

$$i = \frac{60 \times 10}{100} = 6$$

60% percentile = 11.

for 80% .

$$i^0 = \frac{80 \times 10}{100} = 8$$

i.e g

MCQ2
1-5

⑦ MCQ2
 $\rightarrow [16.5, 13.5, 14.5, 8, 15, 9.5, 16.5, 14.5, 6, 14]$

(a) 40% $i = \frac{40 \times 10}{100} = 4$

i.e 8

(b) 60% $i = \frac{60 \times 10}{100} = 6$

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(c) 80% $i = \frac{80 \times 10}{20} = 8$

$i.e. 14.9$

M.C.Q.s

[9, 8, 7, 11, 9, 9, 11, 10.5, 10, 7.5].

(a) 40% $i = \frac{40 \times 10}{100} = 4$ [ie 11]

(b) 60% $i = \frac{60 \times 10}{100} = 6$ [i.e 9]

(c) 80 $i = \frac{80 \times 10}{100} = 8$ [i.e. 10]

(3) Find covariance & correlation of the given data points.

$$a = [2, 4, 5, 6, 7, 8, 9, 10]$$

$$b = [55, 60, 65, 70, 75, 80, 85, 90]$$

$$\text{cov} = \sum_{i=1}^n \frac{(a_i - \bar{a})(b_i - \bar{b})}{n}$$

$$\bar{a} = \frac{\quad}{8} = 6.375$$

$$\bar{b} = \frac{\quad}{8} = 72.5$$

$$\text{cov}(a, b) = \frac{1}{8} \sum (a_i - 6.375)(b_i - 72.5)$$

a_i	$a_i - \bar{a}$	b_i	$b_i - \bar{b}$	$(a_i - \bar{a})(b_i - \bar{b})$	$(a_i - \bar{a})^2$	$(b_i - \bar{b})^2$
2	-4.375	55	-17.5	76.5625	19.1406	306.25
4	-2.375	60	-12.5	29.6875	5.6406	156.25
5	-1.375	65	-7.5	10.3125	1.8906	56.25
6	-0.375	70	-2.5	0.9375	0.1406	6.25
7	0.625	75	2.5	1.5625	0.3906	6.25
8	1.625	80	7.5	12.1875	2.6406	56.25
9	2.625	85	12.5	32.8125	6.8906	156.25
10	3.625	90	17.5	63.4375	13.1406	306.25
					$\sum (a_i - \bar{a})^2$ $= 2.67$ 49.875	$\sum (b_i - \bar{b})^2$ $= 1044$

Now,

$$\text{Cov}(a, b) = \frac{\sum (a_i - \bar{a})(b_i - \bar{b})}{n-1} = \frac{227.5}{7} = 32.5$$

$$\sigma_a = \sqrt{\frac{\sum (a_i - \bar{a})^2}{n-1}} = \sqrt{\frac{49.875}{7}} = \sqrt{7.125} = 2.67$$

$$\sigma_b = \sqrt{\frac{\sum (b_i - \bar{b})^2}{n-1}} = \sqrt{\frac{1044}{7}} = \sqrt{149.14} \approx 12.21$$

Now,

$$r = \frac{\text{Cov}(a, b)}{\sigma_a \sigma_b} = \frac{32.5}{2.67 \times 12.21} = \frac{32.5}{32.58} \approx 0.997$$

$$a = [0, 50, 100, 150, 200, 250, 300, 350]$$

$$b = [60, 75, 88, 95, 92, 80, 65, 40]$$

Calculate Mean.

$$\bar{a} = \frac{\sum_{i=1}^8 a_i}{n} = \frac{1400}{8} = 175$$

$$\bar{b} = \frac{\sum_{i=1}^8 b_i}{n} = \frac{595}{8} = 74.375$$

#	a_i	$a_i - \bar{a}$	b_i	$b_i - \bar{b}$	$(a_i - \bar{a})(b_i - \bar{b})$	$(a_i - \bar{a})^2$	$(b_i - \bar{b})^2$
	0	-175	60	13.625	-2384.375	30625	206.64
	50	-125	75	2.625	-328.125	15625	6.89
	100	-75	88	13.625	-1021.875	5625	185.64
	150	-25	95	20.625	-515.625	625	310.64
	200	25	92	17.625	440.625	625	310.64
	250	75	80	5.625	421.875	5625	31.89
	300	125	65	-9.375	-1171.875	15625	148.64
	350	175	40	-34.375	-6015.625	30625	1181.64
				Σ	Σ	Σ	Σ
				$= -7429$	$= 1060.57$	$= 14910.71$	$= 2429.87$

Now,

$$\text{Cov}(a, b) = \frac{\sum_{i=1}^n (a_i - \bar{a})(b_i - \bar{b})}{n-1} = \frac{-7429}{7} = -1060.57$$

$$\sigma_a = \sqrt{\frac{\sum_{i=1}^n (a_i - \bar{a})^2}{n-1}} = \sqrt{\frac{14910.71}{7}} \approx 122.12$$

$$\sigma_b = \sqrt{\frac{2429.87}{7}} = \sqrt{347.12} = 18.63$$

$$\begin{aligned} \therefore r &= \frac{\text{Cov}(a, b)}{\sigma_a \sigma_b} \\ &= \frac{-1060.57}{122.12 \times 18.63} \\ &\approx -0.466 \end{aligned}$$